Project

Data Network Model based on the TCP/IP architecture

Danylo Viakhiriev

1. Designing IP address scheme

Requirements for the network are as follows:

IP address scope: 150.198.144.0/20

Maximal required count of end systems (ie. PC) in each subnet is:

A	511
B	332
C	360
D	413
E	446
F	88
G	131
H	401

To find the subnet mask for each subnet using VLSM, it is necessary to find the smallest number n (representing the number of host bits), such that 2^n - 2(two IPs are reserved for network IP and broadcast) would be bigger than the number of required hosts.

The exact mask is then 32 - n and it is /x notation mask, which can be converted to x.x.x.x notation by binary-decimal conversion, as x.x.x.x notation is 4 numbers each represented by 8 bits, 0-255.

As there are 8 networks in the requirements, there are thus 8 more point-to-point networks I-P between routers that connect networks A-H. Those point-to-point networks then need to have only 2 host IPs.

Including networks I till P(point-to-point networks between routers) and then sorting networks from largest number of hosts till smallest:

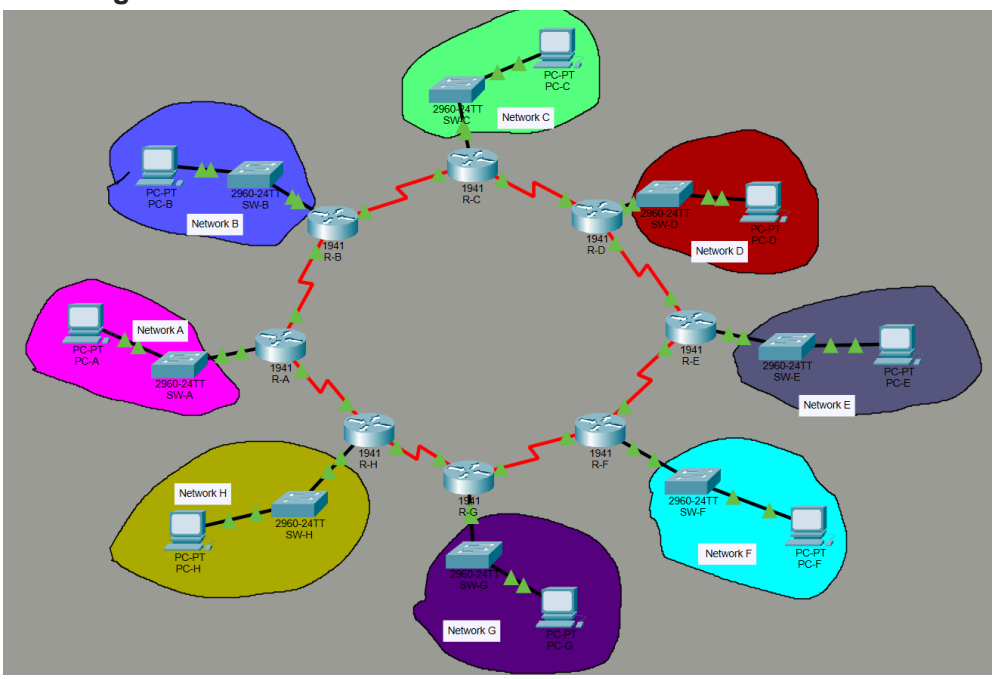
Network	Network IP	Network Mask /x	Network Mask x.x.x.x	Total # of IPs	# of usable IPs	First IP	Last IP	Broadcast IP
A	150.198.144.0	/22	255.255.252.0	1024	1022	150.198.144.1	150.198.147.254	150.198.147.255
E	150.198.148.0	/23	255.255.254.0	512	510	150.198.148.1	150.198.149.254	150.198.149.255
D	150.198.150.0	/23	255.255.254.0	512	510	150.198.150.1	150.198.151.254	150.198.151.255
H	150.198.152.0	/23	255.255.254.0	512	510	150.198.152.1	150.198.153.254	150.198.153.255
C	150.198.154.0	/23	255.255.254.0	512	510	150.198.154.1	150.198.155.254	150.198.155.255
B	150.198.156.0	/23	255.255.254.0	512	510	150.198.156.1	150.198.157.254	150.198.157.255
G	150.198.158.0	/24	255.255.255.0	256	254	150.198.158.1	150.198.158.254	150.198.158.255
F	150.198.159.0	/25	255.255.255.128	128	126	150.198.159.1	150.198.159.126	150.198.159.127
I	150.198.159.128	/30	255.255.255.252	4	2	150.198.159.129	150.198.159.130	150.198.159.131
J	150.198.159.132	/30	255.255.255.252	4	2	150.198.159.133	150.198.159.134	150.198.159.135
K	150.198.159.136	/30	255.255.255.252	4	2	150.198.159.137	150.198.159.138	150.198.159.139
L	150.198.159.140	/30	255.255.255.252	4	2	150.198.159.141	150.198.159.142	150.198.159.143
M	150.198.159.144	/30	255.255.255.252	4	2	150.198.159.145	150.198.159.146	150.198.159.147
N	150.198.159.148	/30	255.255.255.252	4	2	150.198.159.149	150.198.159.150	150.198.159.151
O	150.198.159.152	/30	255.255.255.252	4	2	150.198.159.153	150.198.159.154	150.198.159.155
P	150.198.159.156	/30	255.255.255.252	4	2	150.198.159.157	150.198.159.158	150.198.159.159

Table 1: IP addressing table for the Data Network model meeting task requirements, networks sorted in descending order based on their size

2. Designing the model

Now, it is possible to design the model in Cisco Packet Tracer and assign calculated values to the devices, following required topology of the network and using IP addressing plan based on Table 1.

Figure 1: Logical overview of the created network model in Cisco Packet Tracer



All hosts in each network are represented by only one PC, that will use the first usable IP in its network. Default gateway of each PC is set to the router on its network.
In this configuration, switches do not net any specific configuration, they are just connected to the PC via FastEthernet.
Routers are connected to switches via Gigabit Ethernet cables, and connected between each other via Serial DCE cables with a set clock rate of 64000 on the DCE end.
The IP address for the Gigabit port on the router is the last usable IP address in the appropriate network (e.g. 150.198.147.254 for router R-A). IP addresses for Serial ports are from appropriate networks in I - P range of networks.

Static routing is used, with all the packets travelling clockwise.

Table 2: Assignment of IP addresses for all the routers

Ports / Routers	R-A	R-B	R-C	R-D	R-E	R-F	R-G	R-H
GigEth 0/0	150.198.147.254/22	150.198.157.254/23	150.198.155.254/23	150.198.151.254/23	150.198.149.254/23	150.198.159.126/25	150.198.158.254/24	150.198.153.254/23
Serial 0/1/0	150.198.159.157/30	150.198.159.130/30	150.198.159.134/30	150.198.159.138/30	150.198.159.142/30	150.198.159.146/30	150.198.159.150/30	150.198.159.154/30
Serial 0/1/1	150.198.159.129/30	150.198.159.133/30	150.198.159.137/30	150.198.159.141/30	150.198.159.145/30	150.198.159.149/30	150.198.159.153/30	150.198.159.158/30

Table 3: Assignment of IP addresses for all the PCs (port FastEthernet0)

PC-B	PC-C	PC-D	PC-E	PC-F	PC-G	PC-H
150.198.156.1/23	150.198.154.1/23	150.198.150.1/23	150.198.148.1/23	150.198.159.1/25	150.198.158.1/24	150.198.152.1/23

3. Testing the model

Using the ICMP ping request, it is confirmed that all the devices on the network are mutually accessible (Clockwise static routing can be observed in the Simulation mode).

4. Conclusion

In this project, a fully functional network with set prerequisites for the number of hosts was created, using VLSM while saving IP addresses. Logical model of said network was created in Cisco Packet Tracer and tested with ICMP tool (ping), confirming mutual accessibility of each device.